

Properties of cut-sets Theorem 4:

Ringsum of any 2 cutsets in a graph is either a third set or an edge disjoint union of cutsets.

Proof:

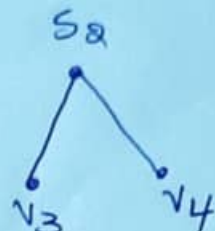
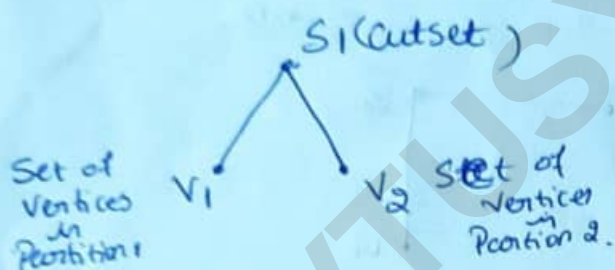
* let G be a connected graph.

* let S_1 & S_2 are two cutsets,

* we know that cutset partition the graph G into 2.

* S_1 Partition the graph into V_1 & V_2

* S_2 Partition the graph into V_3 & V_4 .



$$V_1 \cup V_2 = V \quad (\text{No vertices in } G)$$

$$V_1 \cap V_2 = \phi$$

$$V_3 \cup V_4 = V$$

$$V_3 \cap V_4 = \phi$$

Ringsum

$$V = V_1 \cup V_2$$

E = edges both in either G_1 or G_2 not in both

$$= (E_1 \cup E_2) - (E_1 \cap E_2)$$

$$\text{Let } V_5 = (V_1 \cap V_4) \cup (V_2 \cap V_3) \Rightarrow V_1 \oplus V_3$$

$$V_6 = (V_1 \cap V_3) \cup (V_4 \cap V_2) \Rightarrow V_2 \oplus V_3$$

consider

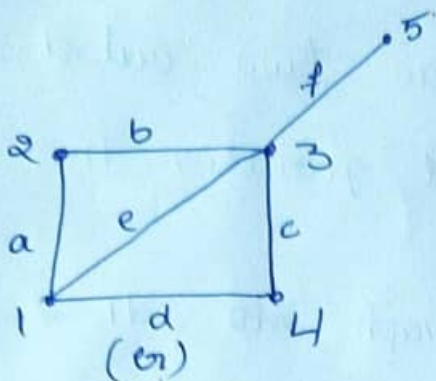
$$S_1 \oplus S_2$$

Edges belonging to this ringsum are only from V_5 to V_6 and also contains all the edges b/w V_5 & V_6 .

*1) Removal of $S_1 \oplus S_2$ makes the graph disconnected
 Partitions the vertex set into V_5 & V_6
 such that $V_5 \cup V_6$ all vertex, $V_5 \cap V_6$ null set,
 Set V empty set

Hence the proof.

Eg:



$$S_1 = \{a, b\}$$

$$S_2 = \{a, e, d\}$$

$$S_1 = \{a, b\}$$

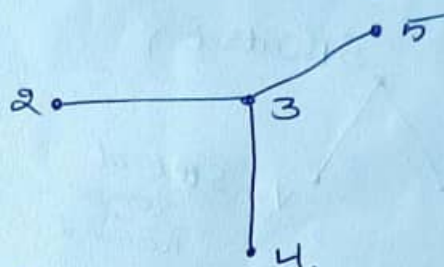
$$S_2 = \{a, e, d\}$$

2.



$$V_1 = \{2\}$$

$$V_2 = \{1, 3, 4, 5\}$$



$$V_3 = \{1\}$$

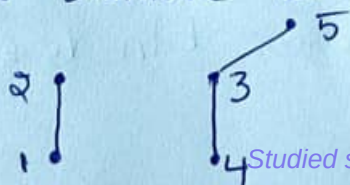
$$V_4 = \{2, 3, 4, 5\}$$

$$V_5 = (V_1 \cap V_4) \cup (V_2 \cap V_3) = \{2\} \cup \{1\} = \{1, 2\}$$

$$V_6 = (V_1 \cap V_3) \cup (V_2 \cap V_4) = \{\} \cup \{3, 4, 5\} = \{3, 4, 5\}$$

$$S_1 \oplus S_3 = \{b, e, d\}$$

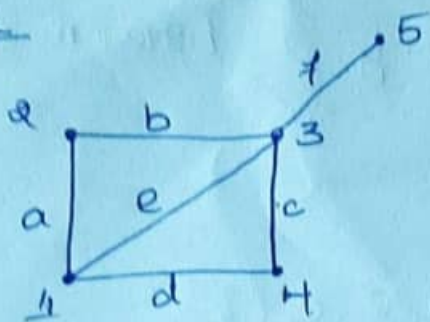
If i remove b, e, d from the graph becomes disconnected



$$V_5 = \{1, 2\}$$

$$V_6 = \{3, 4, 5\}$$

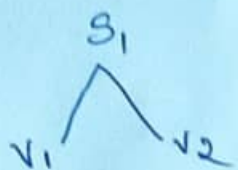
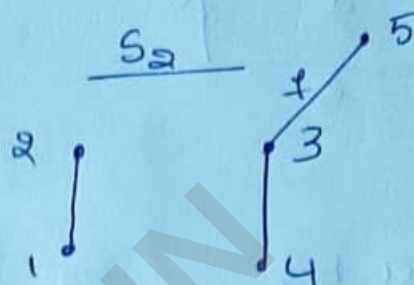
Eg: 2



$$S_1 = \{a, c, e\}$$

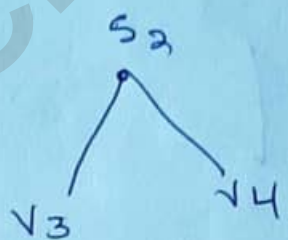
$$S_1 \oplus S_2 = \{a, b, c, d\}$$

$$S_2 = \{b, e, d\}$$



$$v_1 = \{1, 4\}$$

$$v_2 = \{2, 3, 5\}$$



$$v_3 = \{3, 4, 5\}$$

$$v_4 = \{1, 2\}$$

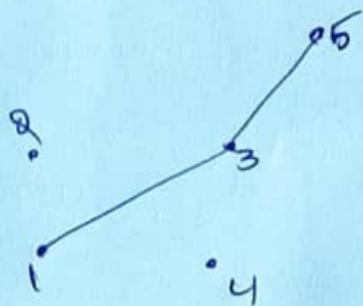
$$v_6 = (v_1 \cap v_3) \cup (v_2 \cap v_4) = \{1\} \cup \{3, 5\} = \{1, 3, 5\}$$

$$v_5 = (v_1 \cap v_4) \cup (v_2 \cap v_3) = \{4\} \cup \{2\} = \{4, 2\}$$

$$S_1 \oplus S_2 = \{a, b, c, d\} = \{a, b\} \cup \{c, d\}$$

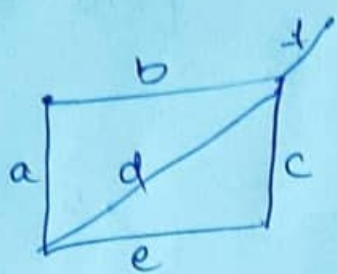
↓

union of edges is
joint of cut set.

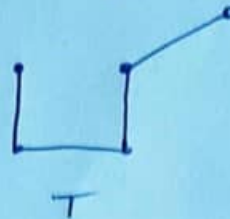


Fundamental cutset:

A cutset containing exactly 1 branch of spanning tree.



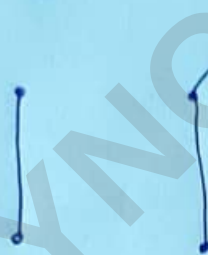
$$T = \{a, e, c, f\}$$



$$S_1 = \{a, b\}$$



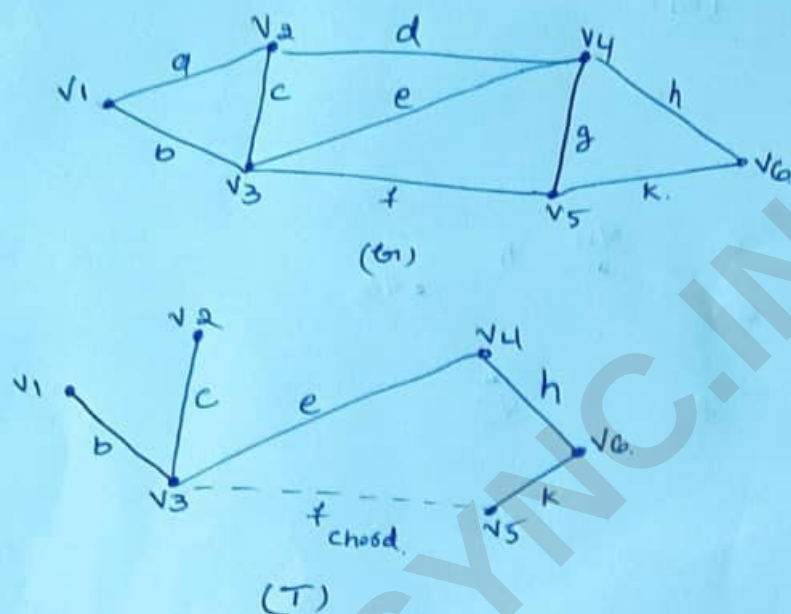
$$S_2 = \{b, d, e\}$$



Theorem 5

- *1 with respect to a given spanning tree T , a chord $\bar{a}$ that determines a fundamental circuit $\Gamma_{\bar{a}}$ occurs in every fundamental cut-set associated with branches in $\bar{a}$ and in no other.

Eg:



- *1 spanning tree $\{b, c, e, h, k\}$

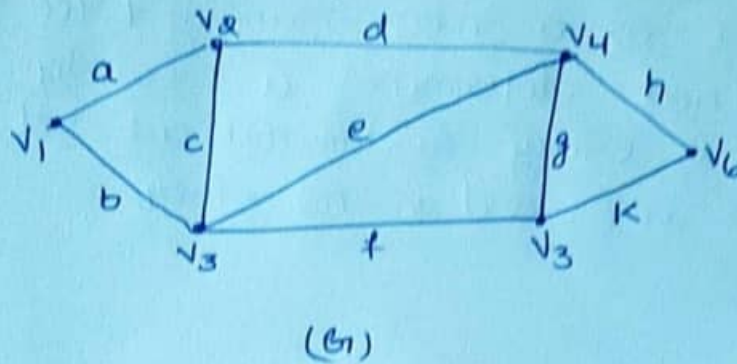
The fundamental circuit made by chord $\bar{f}$ is $\{f, e, h, k\}$

- *1 Three fundamental cutsets determined by the 3 branches e, h, k are.

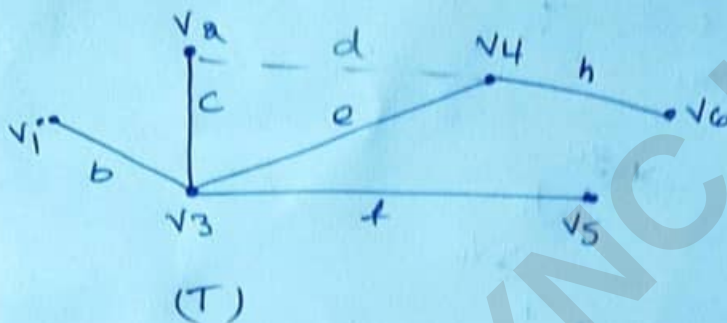
- *1
- | | |
|---------------|---------------------------|
| $\{d, e, f\}$ | cut set determined by e |
| $\{f, g, h\}$ | cut set determined by h |
| $\{f, g, k\}$ | cut set determined by k |

- *1 chord $\bar{f}$ occurs in each of these 3 fundamental cutsets and there are no other fundamental cut-sets that contain $\bar{f}$.

Eg: a Fundamental cutsets



Spanning Tree:-



$\{c, e, d\}$ - circuit

cut-set

$\{a, c, d\}$ - fundamental

$\{d, e, f\}$ - fundamental

$\{e, f, g\}$ - cut-set

$\{f, g, k\}$ - fundamental

cut set

$\{a, \overline{b}\}$ - fundamental cutset

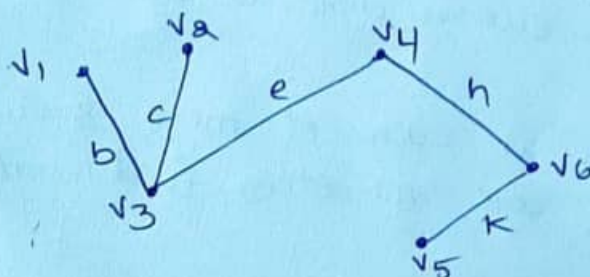
$\{a, \overline{c}, d\}$ - fundamental cutset

$\{d, e, f\}$ - not fundamental cutset

$\{f, g, h\}$ - not fundamental cutset

$\{f, g, k\}$ - fundamental cutset

Spanning Tree:-



fundamental cutset

$\{a, \underline{b}\}$

$\{a, \underline{c}, d\}$

$\{d, \underline{e}, f\}$

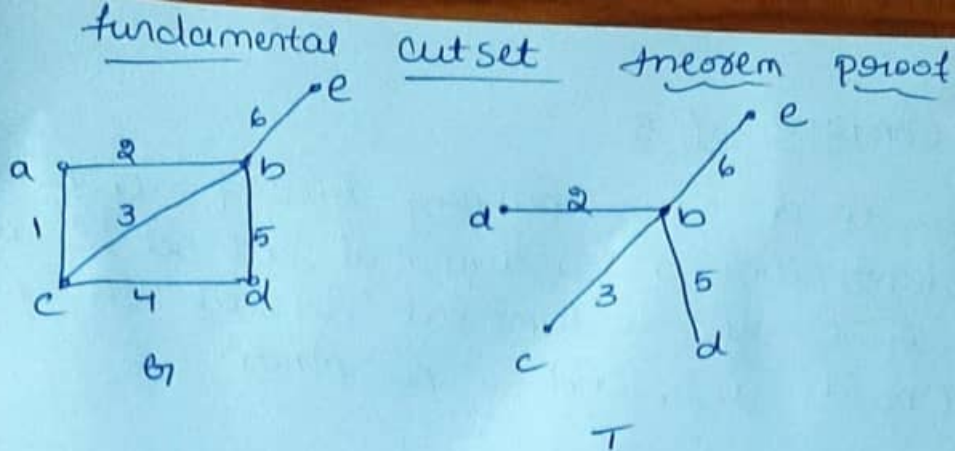
$\{f, \underline{g}, h\}$

$\{f, \underline{g}, k\}$

only on
branch
is
each cutset

so fundamental cutset

Fig:



1) Spanning tree $\{2, 3, 5, 6\}$

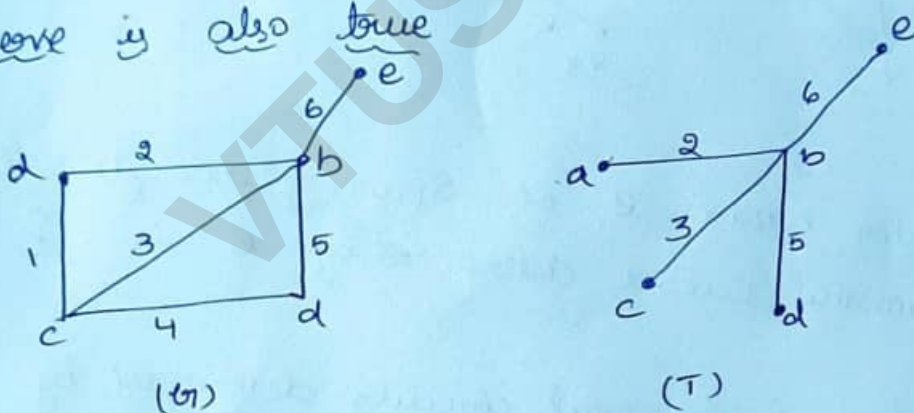
*1 adding a chord 4 gives fundamental circuit $\{3, 5, 4\}$
↑
chord

*2/ fundamental cut set determined by 3, 5 are

fundamental cut set determined by 3: $\{2, 3, 4\}$ ↑
branch can't be fundamental cut set
 fundamental cut set determined by 5: $\{4, 5\}$

Chord '4' appears in fundamental cut set not in other.

Converse is also true



fundamental cut-set determined by '3' is $\{1, 3, 4\}$ ↑
branch

fundamental circuit formed by Chord 1 and Chord 4

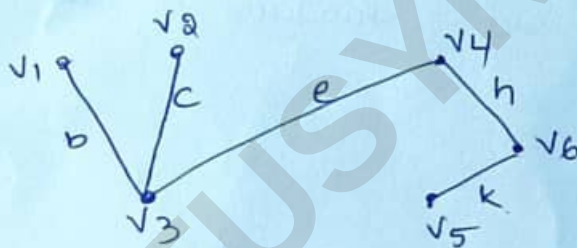
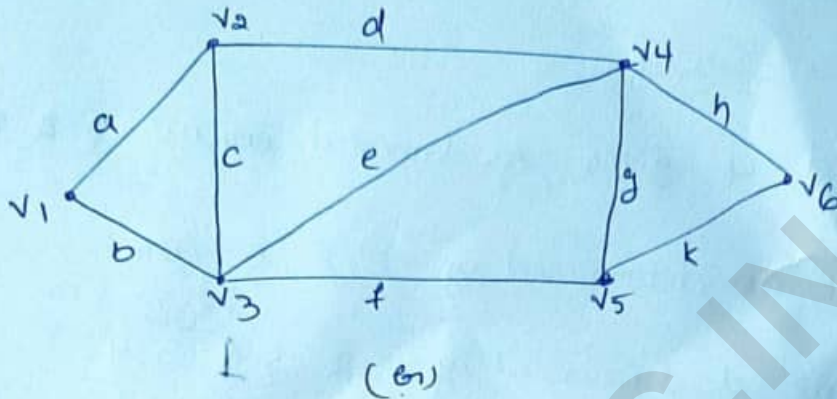
fundamental circuit formed by Chord 1: $\{1, 2, 3\}$

fundamental circuit formed by Chord 4: $\{3, 5, 4\}$

branch 3 is present in all the fundamental circuit determined by other branches in cutset, not in any other.

Theorem 6 : converse of 5

With respect to a given spanning tree T , a branch b_i that determines a fundamental cut-set S is contained in every fundamental circuit associated with the chords in S , and in no others.



*1 Consider branch e of spanning tree $\{b, c, e, h, k\}$
fundamental cutset determined by e is $\{e, d, f\}$

*1 The two fundamental circuits determined by chord d & f are.

F.c determined by chord d : $\{d, c, e\}$

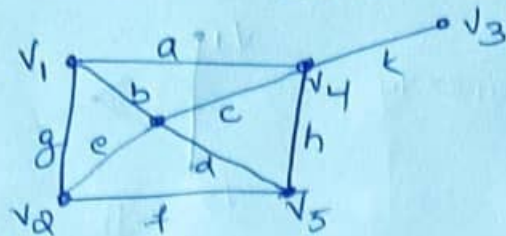
F.c determined by Chord f : $\{f, e, h, k\}$

*1 Branch e is contained in both fundamental circuits.

Edge connectivity:

* Each cut-set of a connected graph G consists of a certain NO of edges. NO of edges in the smallest cut-set is defined as the edge connectivity.

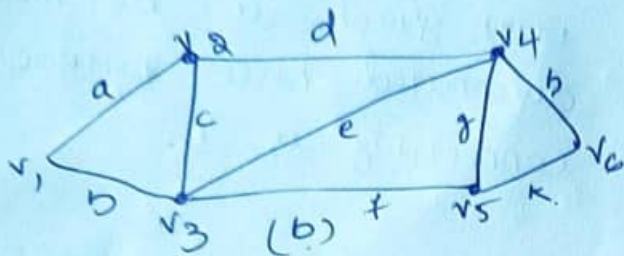
Eg:



(a)

Smallest cut-set for given graph (a) is $\{k\}$.

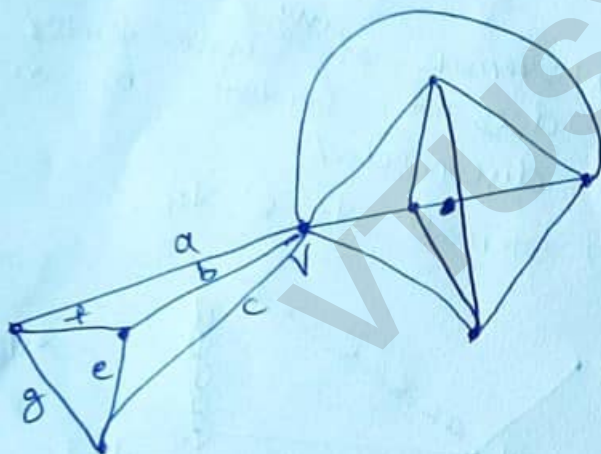
$\therefore$ Edge connectivity is 1.



(b)

Smallest cut-set for given graph (b) is $\{a, b\}$, $\{h, k\}$

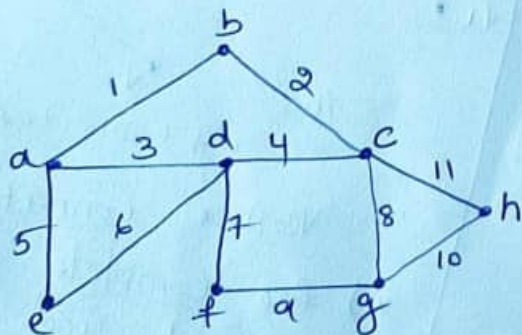
$\therefore$ edge connectivity is 2



(c)

Smallest cut-set for given graph (c) is $\{a, b, c\}$, $\{a, f, g\}$

$\therefore$ edge connectivity is 3.



(d)

* edge connectivity of graph (d) is 2

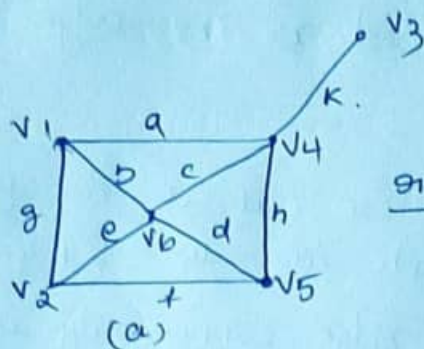
If we remove edge $\{1, 2\}$ or $\{11, 10\}$ or $\{7, 9\}$ graph becomes disconnected

* If we remove vertex $\{d, c\}$ or $\{a, d\}$ or $\{d, g\}$ then graph becomes disconnected $\therefore$

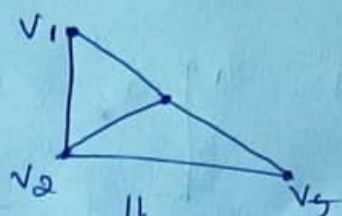
Vertex connectivity:

The minimum number of vertices whose removal from 'G' leaves the remaining graph disconnected.

Eg:



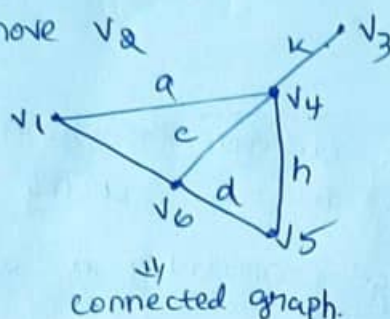
remove vertex v_4



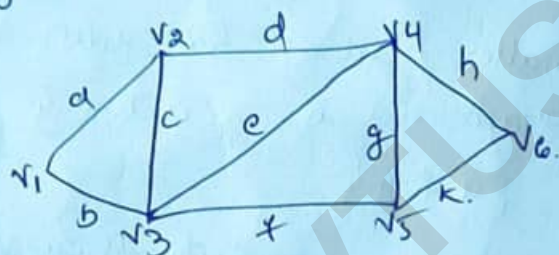
disconnected graph.

$\therefore$ if we remove v_4 our graph (a) becomes disconnected. hence vertex connectivity is 1.

remove v_2



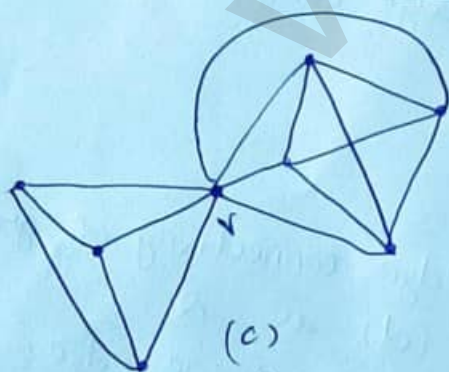
Eg:



* remove of ^{any} single vertex does not convert (b) into disconnected

* remove v_2 & v_4 .

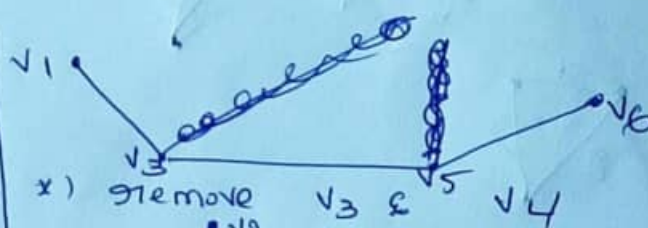
Eg:



remove ' v ' disconnect graph



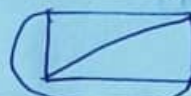
$\therefore$ vertex connectivity is 1.



(b) disconnected

$\therefore$ vertex connectivity is 2

* for complete graph we can't find vertex connectivity



$K.C = 3$

* edge connectivity is 3
6 v.c is defined for connected graph

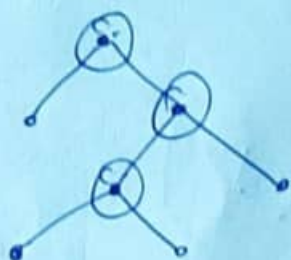
Separable graph

a graph is said to be separable if there exist a subgraph g in such a way that g & complement of g have 1 common vertex.

or

a graph which has only ~~one~~ vertex connectivity 1 then they are called separable graph.

tree;



In a tree every vertices which has degree greater than one are called cut-vertices in tree.

Cut-vertices: If we have a graph G and removal of an vertex or vertices makes the G disconnected then those vertex or vertices are called cut-vertices.

Theorem: The edge connectivity of a graph can't exceed the degree of the vertex with smallest degree in G .

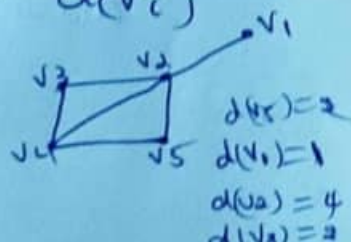
Proof: let v_i is the vertex which has smallest degree in G .

$v_i \quad d(v_i) \rightarrow$ Smallest deg in G .

Removal of edges associated with v_i disconnect G
cut-set

$\therefore$ There exist a cut-set, no. of edges in that cutset is less than or equal to $d(v_i)$

$\therefore$ edge connectivity is $\leq d(v_i)$

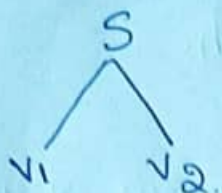


Theorem: The vertex connectivity of any graph can never exceed the edge connectivity of G .

Proof: Let α be edge connectivity of G

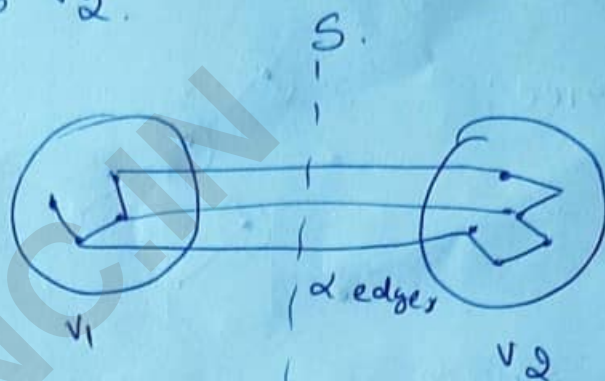
* Take cut set S

* S partition the vertex set of G into V_1 & V_2



* α edges are connected to vertices in V_1 & V_2 .

vertex connectivity of graph $\leq \alpha$.



* α edges are connected to α vertices in V_1 & α vertices in V_2

* If I remove α vertex in any set V_1 or V_2 this will disconnect the graph.

Thus vertex connectivity must be $\leq \alpha$.

$$\begin{aligned} \nu_c &\leq \text{edge connectivity of graph } G \\ &\leq \frac{2e}{n} \text{ in connected graph.} \end{aligned}$$



$$\left\lfloor \frac{2e}{n} \right\rfloor$$

Maximum vertex connectivity

— Integral part of

e — no of edges connected graph.

n — no of vertex of connected graph